

2019 Troubleshooting Guide for DJI Agras Products

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1. T16/MG-1P Aircraft firmware update failure

- Issue: The T16/MG-1P aircraft firmware update fails when using DJI Assistant 2 for MG, and an error code is shown on the software.
- Aircraft model: T16, MG-1P, and MG-1P RTK
- Troubleshooting:

Aircraft	Code	Defective Module	Solution
T16/MG-1P	0x02a303	Flight Controller Loader	Replace the aerial-electronics system board/main controller.
T16/MG-1P	0x02c303	Flight Controller Loader	Replace the aerial-electronics system board/main controller.
T16/MG-1P	0x020b03	Battery	Replace the battery.
T16/MG-1P	0x020c03	M1 ESC	Replace the M1 ESC.
T16/MG-1P	0x022c03	M2 ESC	Replace the M2 ESC.
T16/MG-1P	0x024c03	M3 ESC	Replace the M3 ESC.
T16/MG-1P	0x026c03	M4 ESC	Replace the M4 ESC.
T16/MG-1P	0x028c03	M5 ESC	Replace the M5 ESC.
T16/MG-1P	0x02ac03	M6 ESC	Replace the M6 ESC.
T16/MG-1P	0x021803	Radar ESC	Replace the radar.
T16/MG-1P	0x023803	Radar Wireless Ground Unit	Replace the radar.
T16/MG-1P	0x027803	Radar	Replace the radar.
T16/MG-1P	0x029803	Radar Wireless Air Unit	Replace the radar.
T16/MG-1P	0x021c03	Central Board	Replace the spraying board /central board.
T16/MG-1P	0x022803	1860 Chip	Replace the aerial-electronics system board/main controller.
T16/MG-1P	0x02e903	SDR Module	Replace the aerial-electronics system board/SDR module.
T16	0x02fa03	RTK Board	Replace the aerial-electronics system board.

DJI Agras

T16	0x021e03	No.1 Delivery Pump ESC	Replace the No.1 delivery pump ESC.
T16	0x023e03	No.2 Delivery Pump ESC	Replace the No.2 delivery pump ESC.
T16	0x025e03	No.3 Delivery Pump ESC	Replace the No.3 delivery pump ESC.
T16	0x027e03	No.4 Delivery Pump ESC	Replace the No.4 delivery pump ESC.
T16	0x022503	Flow Meter	Replace the flow meter.
MG-1P	0x02cc03	M7 ESC	Replace the M7 ESC.
MG-1P	0x02ec03	M8 ESC	Replace the M8 ESC.
MG-1P	0x02da03	D-RTK 2	Replace the D-RTK 2.
MG-1P	0x021a03	GPS Module	Replace the GPS module.
MG-1P	0x029e03	No.1 Delivery Pump ESC	Replace the central board.
MG-1P	0x02be03	No.2 Delivery Pump ESC	Replace the central board.
MG-1P	0x02fe03	Spreader ESC	Replace the spreader ESC.
MG-1P	0x024503	Spreader Control Board	Replace the spreader control board.

2. MG series drone main controller firmware update failure

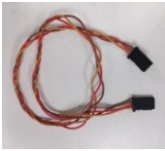
- Issue: After the main controller firmware update fails several times, the eight ESC LEDs blink yellow.
- Aircraft model: MG serial products
- Solution:
 1. Power off the drone and disconnect the cables on the main controller, except for the cable connecting the main controller and PMU (for MG-1 and MG-1S) or the cable connecting the main controller and central board (for MG-1A and MG-1P). These two cables are used for supplying power to the main controller.
 2. Connect the 3-pin Dupont cable or three Dupont cables to the corresponding port as shown below.
 3. Power on the drone, wait for five seconds, and then power off the drone.
 4. Disconnect the Dupont cable(s) and attach the previously disconnected cables back to the main controller.
 5. Power on the drone and update the firmware again. The firmware will possibly update successfully, but if it still fails, it is recommended to replace the main controller.
 6. This method is not suitable for aircraft that have hit high-voltage lines.



MG-1



MG-1S/MG-1A/MG-1P

Product	Port	Tool 1	Tool 2
MG-1	F1 and F2		
MG-1S	ESC Ports M7 and M8		
MG-1A	ESC Ports M7 and M8		
MG-1P	ESC Ports M7 and M8		
		3-pin Dupont cable	Dupont cable x3

3. Uploading the field file to the DJI Agriculture Management Platform fails, times out, or gets stuck

- Issue: Uploading the field file to the DJI Agriculture Management Platform fails, times out, or gets stuck.
- Aircraft model: All
- Troubleshooting

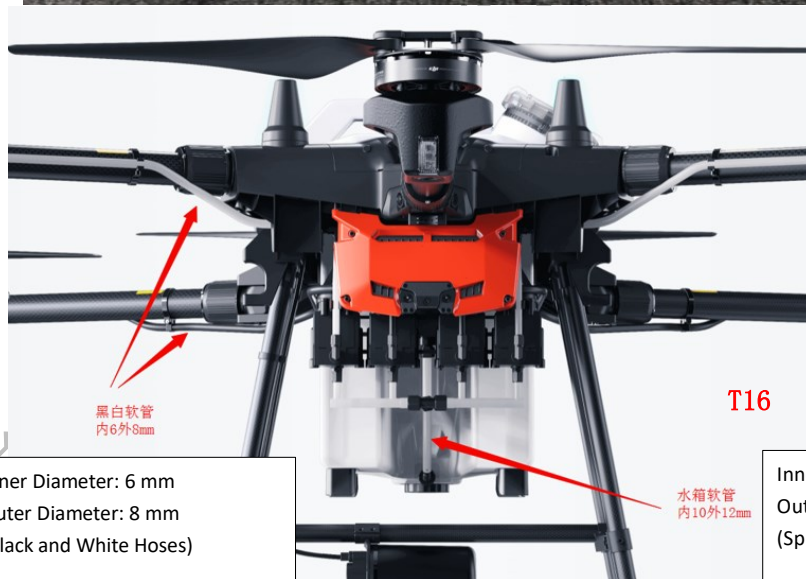
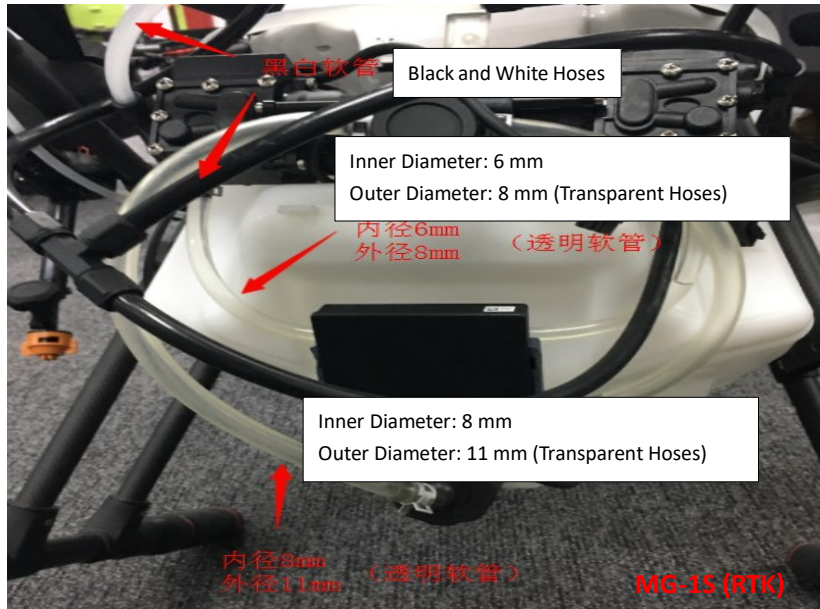
Issue	Troubleshooting
Uploading fails	1. Upload the field file again. 2. Check the network environment and see if web pages can be browsed normally. 3. Check whether the field file is already uploaded to the DJI Agriculture Management Platform, but the DJI Agriculture Management Platform prompts that it has not be uploaded. 4. If the field file is not uploaded to the DJI Agriculture Management Platform while the remote controller has field information, please upload the task again by connecting to a mobile phone hotspot or other Wi-Fi network.
Uploading times out	

	5. If the issue persists, record the customer account, password, and QR code, and then send the information to DJI Support.
Uploading gets stuck	1. Turn off the app or restart the remote controller. 2. Check the network environment to determine whether the issue is caused by an unstable network environment. If yes, please upload the file again under normal network conditions. 3. If not, the issue may occur because the amount of information to be synchronized is too large. Please upload it in sections. 4. If the issue occurs frequently, please check if the remote controller firmware or app version is the latest. It is recommended that the user updates the firmware and app to the latest version. 5. If the issue persists, record the customer account, password, and QR code, and then send the information to DJI Support.

4. DJI Agras drone hoses specifications comparison

Aircraft Type	Model	MG-1	MG-1S (RTK)	MG-1S Advanced MG-1P (RTK)	T16
Delivery Pump->Nozzle (White Hoses)		—	Inner Diameter: 6 mm Outer Diameter: 8 mm Nylon (PA) 1.2 MPa		
Delivery Pump->Nozzle (Black Hoses)					
Spray Tank Hoses		—			Inner Diameter: 10 mm Outer Diameter: 12 mm PU
Spray Tank->Flow Meter (Transparent Hoses)		—	Inner Diameter: 8 mm Outer Diameter: 11 mm PU	—	
Flow Meter-> Secondary Spray Tank (Transparent Hoses)					
Secondary Spray Tank->Delivery Pump (Transparent Hoses)		—	Inner Diameter: 6 mm Outer Diameter: 8 mm PU	—	
Transparent Hoses		Inner Diameter: 8 mm Outer Diameter: 11 mm TPU	—		
Material of Spray Tank		HDPE			

Refer to the following two pictures for the specific locations



5. DJI Agras drone activation process

5.1 Activating the MG-1S/MG-1S RTK

- For aircraft firmware v3.1.0.0 or earlier, first activate the drone on the dealer management platform, and then activate it with the DJI MG app. For more details, watch the tutorial video: [xxxxxx](#)
- For firmware v3.1.0.0 or later, activate the aircraft with the DJI MG app. For more details, watch the tutorial video: [xxxxxx](#)

批注 [k1]: 添加视频英文链接

5.2 Activating the MG-1S Advanced, MG-1P, and MG-1P RTK

- Activate the aircraft with the DJI MG app. For more details, watch the tutorial video: [xxxxxx](#)

5.3 Activating the T16

- Activate the aircraft firmware with the DJI MG app. For more details, watch the tutorial video: [xxxxxx](#)

6. An “IMU 10” error code is shown on the app

- Issue: An “IMU 10” error code is shown on the app.
- Aircraft model: MG-1S, MG-1S Advanced, MG-1P, MG-1P RTK, and Agriculture Solution Package
- Troubleshooting
 1. Check whether the compass on the landing gear is not connected well, and then calibrate the compass.
 2. If the calibration fails, try to replace the compass and GPS module, and then calibrate the

compass again.

3. If the compass is calibrated successfully and the issue persists, calibrate the IMU.
4. If the issue still cannot be fixed, replace the main controller.

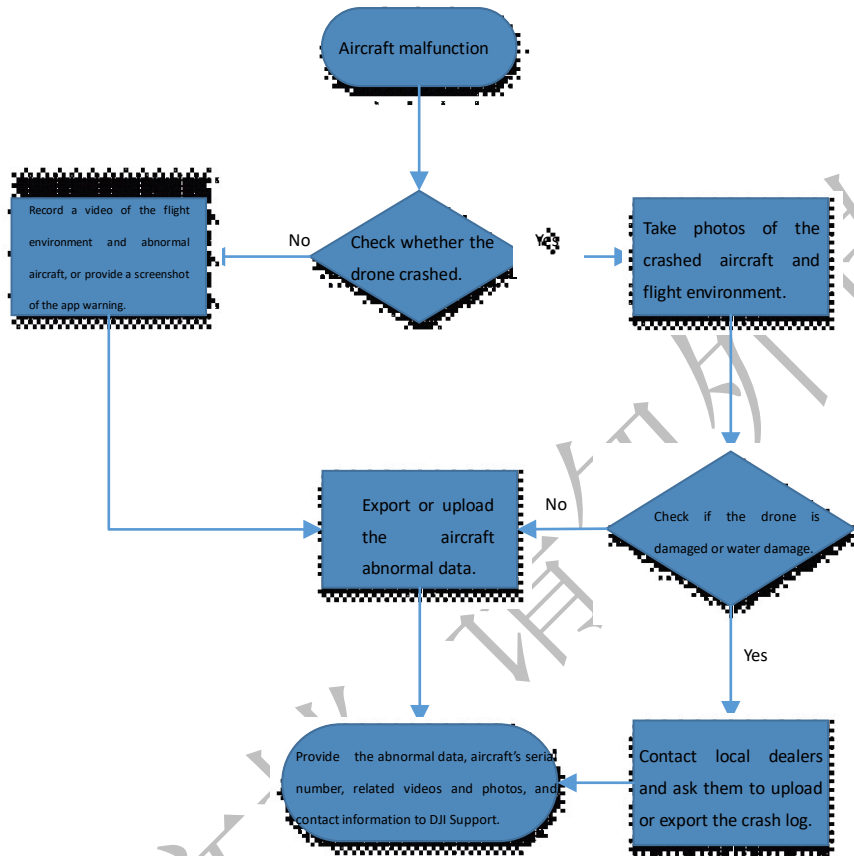
7. Solutions to delivery pump issues

Issue	Aircraft	Cause	Solution
Empty tank warning	MG-1	There is an error with the delivery pump's current data.	Replace the delivery pump.
Delivery pump leakage	MG-1 MG-1S MG-1P T16	The diaphragm and external shell are damaged.	Replace the delivery pump.
		The sealing of the delivery pump connector is poor.	Tighten the hoses and ensure that the air tightness of the connection between the delivery pump and hoses is good.
Delivery pump is disconnected/Delivery pump motor is not detected.	MG-1S MG-1P T16	1. The delivery pump connector is not attached properly. 2. There is an error with the firmware. 3. The delivery pump or central board is damaged.	1. Power off the drone, then remove and insert the delivery pump connector again. Use anhydrous alcohol to clean the connector, and make sure that there is no debris on the connector. 2. Try using a normally functioning delivery pump to see whether or not there is an error with the delivery pump connector or central board connector. Then replace the delivery pump or central board. 3. Check whether the issue occurs after the firmware update. If not, update the firmware to the latest version.
Delivery pump is idling and does not draw water.	MG-1 MG-1S MG-1P T16	1. The delivery pump is empty, and no internal sealing can be formed. 2. The delivery pump is stuck.	1. Disconnect the hoses and use a syringe or pipette to fill the pump and keep water in the pump. 2. Do not use powder or creamy chemicals, which may cause the pump to clog. 3. If the issue persists, there may be an error with the diaphragm. Replace the delivery pump.
Delivery pump does not work and makes abnormal sounds (DDD).	MG-1	1. The delivery pump ESC 4-pin cable is not connected well. 2. The delivery pump's connector on the central board is malfunctioning. 3. The delivery pump ESC is malfunctioning.	1. Ensure that the delivery pump ESC 4-pin cable is connected well. 2. Ensure that the delivery pump power cable is connected well. 3. Replace the delivery pump ESC.

The delivery pump ESC works abnormally.	MG-1S MG-1P	The delivery pump ESC of the MG-1S or MG-1P is on the central board, so there may be an error with the central board's firmware or link.	1. Check whether the issue occurs after the firmware update. If not, update the firmware to the latest version. 2. Check if there are any warnings shown, like the radar connection being abnormal. If so, update the firmware again.
	T16	The delivery pump ESC of the T16 is inside the delivery pump, so there may be an error with the firmware or the hardware.	1. Check whether the issue occurs after the firmware update. If not, update the firmware to the latest version. 2. Try using a normally functioning delivery pump or a delivery pump cable to check whether or not the delivery pump or cable is damaged. Then replace the corresponding component.
The delivery pump pressure sensor works abnormally.	MG-1S MG-1P T16	1. The delivery pump is clogged. 2. The delivery pump's outlet sealing is poor. 3. The pressure sensor is damaged. 4. The liquid is too thick.	1. Do not use powder or creamy chemicals, which may cause the pump to clog. 2. Check whether the delivery pump's outlet and hoses are damaged. 3. Update the firmware to the latest version. 4. Check the pressure value. If the data is abnormal, replace the delivery pump.

8. Solutions to aircraft malfunction issue

- Issue: The aircraft is malfunctioning, which may affect operations.
- Aircraft model: All
- Solution:



9. The screw is stripped during disassembly or assembly.

- The screw is stripped as the screw has been unsecure and tightened many times or the screw and screwdriver do not match.
- Aircraft model: All
- Solution: Use the screw extractor to remove the screw.

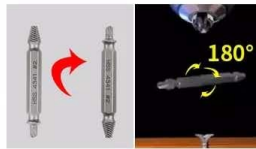
How to use the screw extractor (an electric drill is required):

取出器使用方法

(需配合电钻使用)



1. 先用一头开孔



2. 更换至另外一头



3. 带螺纹一头逆时针取出

1. Pound one side of the screw extractor into the head of the screw.

2. Pound the other side of the screw extractor into the head of the screw.

3. Use a drill to turn the extractor counterclockwise.

10. It is hard to separate the remote controller's front shell and rear shell during repair.

- Issue: There is a buckle inside the remote controller's front shell and rear shell, and it is difficult to separate them during the repair.
- Aircraft model: T16 and MG-1P series
- Solution:
 1. Remove the eight screws on the remote controller's rear shell.
 2. Press the middle of the remote controller rear shell (there is a buckle inside) down vertically with the right hand thumb, and hold the handle with the other four fingers. Hold the screen bracket with the left hand and press down the right thumb with the left thumb. Then pull the handle to the right to separate the front and rear shell. Do not press down on the front shell.
 3. Gently pull the handle, or the flat cable may be damaged.



Flat Cable

排线

卡扣

Buckle

11. The T16 spraying volume is not accurate, or calibration fails

- Issue: The rotation speed of the T16 delivery pump is slow or abnormally/The delivery pump calibration is failed.
- Aircraft model: T16
- Solution:
 1. Before the calibration, open the pressure relief valves to vent out all the air.
 2. Ensure the pressure relief valves are closed during the calibration. If a pressure relief valve is opened, the water discharge capacity of the pump will increase, and the rotation speed will be very low.
 3. Check whether the spray tank is mounted in place, and check if there is air in the pipelines during spraying.
 4. For more details about flow calibration, watch the video at :
<https://cdn.djivideos.com/watch/ca90f3c7-68f4-49b5-9199-b7ffab0a0c2f?autoplay=false&poster=http://videocaption.djicdn.com/poster/20190313/6f7b889ac00598866ab5ad26903e8990.jpg@!1200>
 5. If the issue persists, link the remote controller with the drone, and send the QR code shown in the DJI MG2 app to DJI Support for diagnosis.

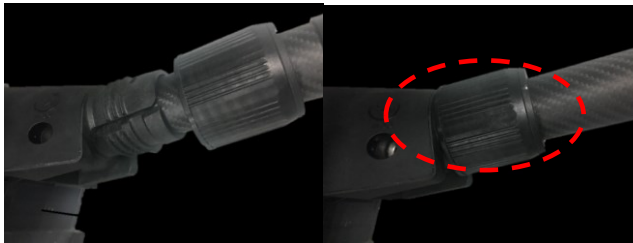
批注 [k2]: 更换英文链接

12. Notices for replacing the T16 propellers during repair

- The propellers have been matched in pairs before being shipped out. It is necessary to replace the propellers in pairs during repair. Do not replace a single propeller individually.

13. The T16 may crash if the sleeve on the aircraft arm is not tightened

- Issue: The T16 may crash if the sleeve on the aircraft arm is not tightened.
- Aircraft model: T16
- Solution: It is recommended to attach the arm sleeve until the thread is completely covered. Then carefully check the tightness of the sleeve. If the sleeve is loose, the drone may crash.



14. There is no FPV image transmission on the app

- Issue: There is no FPV image transmission on the app of the MG-1P remote controller.
- Aircraft model: MG-1P
- Version: V2.1.0009
- Solution:
 1. Check whether the drone used is a MG-1P drone, then go to Remote Controller Settings, Android Settings, and Main Screen, then select DJI MG and restart the remote controller.

批注 [k3]: 以 app 实际英文字段为准



3



2. If the issue persists, link the remote controller with the drone, and send the QR code shown in the DJI MG2 app to DJI Support for diagnosis.

15. When pushing the control sticks to the inner or outer corners, the motors stop

- The Combination Stick Command (CSC) was added for the drone to stop the motors, as the Agras drone operates in a more complicated environment. But the users may wrongly perform the CSC, which causes the aircraft to crash.
- Aircraft model: All
- Only perform the CSC in emergency situations when the aircraft is airborne.



16. Inserting the battery into the T16 improperly may cause the aircraft to crash

- Issue: The aircraft may crash if the battery is not mounted in place.
- Aircraft model: T16
- Note: When a click sound is heard, it indicates that the battery is mounted in place.



There is a buckle in the red circle.

17. There is air in the spraying system or the calibration fails if the spray tank is not mounted in place

- Issue: There is air in the spraying system or the calibration fails if the spray tank is not mounted in place.
- Aircraft model: T16
- Notices:
 1. After the spray tank is placed in the back of the drone, hold the spray tank (red circle areas) with both hands and press the spray tank down lightly until it is tightened.
 2. If the spray tank is not mounted in place, there will be bubbles when pumping.



18. An ESC error prompt is shown on the app

- Issue: The app shows an ESC error prompt.
- Aircraft model: T16
- Notices:
 1. Check whether the motor is misshapen or cannot rotate smoothly. If yes, replace the motor.
 2. Check whether the ESC signal cable is connected well. If not, remove and attach the cable again.
 3. Check whether the issue occurs every time after the drone and remote controller are restarted.
 4. If it does occur every time, guide the users to contact their local dealers for a firmware update.
 5. If the issue occurs occasionally, refresh the firmware or update the firmware to the latest version.
 6. If the issue persists, link the remote controller with the drone again, and send the QR code shown in the app to DJI Support for diagnosis.

19. A “RTK dual antennas not ready” warning is shown on the app

- Issue: A “RTK dual antennas not ready” warning is shown on the app.
- Aircraft model: T16
- Notices:
 1. Make sure that the RTK signal or GPS signal is strong when operating in an open area.
 2. Disconnect and reattach the RTK feeder connector which connects the aerial-electronics board.
 3. Update the firmware to the latest version.
 4. Check whether antenna 1 and antenna 2 can search satellites normally. If the satellite count is 0, replace the RTK feeder and RTK antenna. (Stand at the end of the drone and face towards the direction of aircraft’s nose. The antenna 1 is on the right side, and the antenna 2 is on the left side.)



5. Guide the customers to contact their dealers and replace the RTK antenna to see if the issue is fixed. If not, replace the RTK feeder.
6. Replace the RTK antenna, RTK feeder, and aerial-electronics board to see if the issue is fixed.
7. If the hardware is working normally, update the firmware to the latest version.
8. If the issue persists, link the remote controller with the drone, and send the QR code shown in the DJI MG2 app to DJI Support for diagnosis.

20. There is an error with the T16 liquid level meter

If (1) a prompt in the app states that there is liquid in the tank, but actually the tank is empty; or (2) a warning states that the spray tank is empty is shown on the app, but actually there is liquid in the tank, perform the following troubleshooting steps:

A prompt stating that there is liquid in the tank is shown on the app, but actually the tank is empty

1. Update the aircraft's firmware to the latest version.
2. Calibrate the flow meter first, then calibrate the delivery pump.
3. Check whether "Initial Pesticide(L)" is set in the app (if the initial dose is larger than the liquid filled in, there will be no warnings shown).
4. Check whether the liquid level meter and liquid level meter sensor are functioning.

Inspecting the liquid level meter

1. Replace the device and perform the test. If the liquid is below the upper and lower liquid level meters, the "Remaining Pesticide" will be shown as "<1.3L". If the liquid is above the lower liquid level meter but below the upper liquid level meter, the "Remaining Pesticide" will be shown as "8.5L>Remaining Pesticide>1.3L". If the liquid is above the upper liquid level meter, "Remaining Pesticide" will be shown as ">8.5L".
2. Add water or use a hand to toggle the liquid level meter and check the "Remaining Pesticide" shown in the app to see if the liquid level meter is functioning.
3. If the tank is empty and the "Remaining Pesticide" shown in the app is ">1.3L", the lower liquid level meter's magnetic pole of the magnet is reversed. If the tank is fully filled with the water and the "Remaining Pesticide" shown in the app is "<8.5L", the upper liquid level meter's magnetic pole of the magnet is reversed. If either of these situations occurs, then replace the corresponding upper or lower liquid level meter.

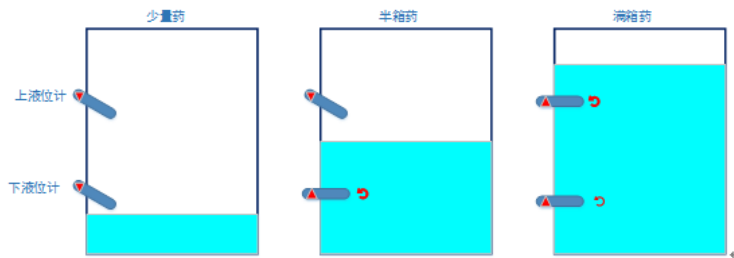


图 1.3 液位计检测药量示意图

Inspecting the Liquid Level Meter Sensor

1. Mount the spray tank to another drone and check if the issue is fixed. If yes, adjust the lower sensor's installation position.
2. Adjust the sensor's position up or down to see if the issue is fixed.
3. If not, replace the liquid level meter's cable.
4. If the issue persists, link the remote controller with the drone and send the QR code shown in the app to DJI Support for diagnosis.



A warning stating that the spray tank is empty is shown on the app, but actually there is liquid in the tank

1. Check whether the upper and lower liquid level meters are working normally.
2. Tell the users to set the Initial Pesticide value based on the actual pesticide filled in the tank.

21.The drone's flight speed is slow or the radar makes a wrong report (causing the drone to enter the obstacle avoidance mode)

- Issue: The drone's flight speed is slow or the radar makes a wrong report (causing the drone to enter the obstacle avoidance mode). After troubleshooting, it is found that the radar angle is tilted.
- Aircraft model: T16
- Notices:
 1. Ensure that the radar is mounted properly (face towards the tail of the drone to check if the radar port faces the right landing gear).
 2. Guide the user to remove the radar and mount it again.
 3. Perform the radar calibration with the DJI MG2 app.
 4. If the issue persists, link the remote controller with the drone and send the QR code shown in the app to DJI Support for diagnosis (provide a video of the issue if necessary).

22. The T16 delivery pump is blocked

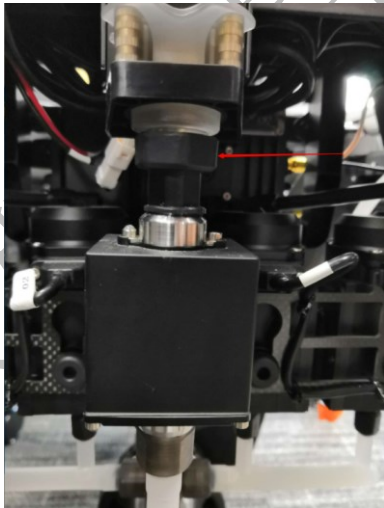
If the nozzle or delivery pump is blocked or the hoses are folded, a warning stating that the delivery pump is blocked will be shown on the app.

Check/Clean Nozzles

1. Check whether the hoses are folded or bent.
2. Calibrate the liquid level meter or delivery pump.
3. Use a new delivery pump to check whether the issue is fixed (if the delivery pump is out of warranty, replace the diaphragm).
4. If the issue persists, link the remote controller with the drone, and send the QR code shown in the app and spraying video to DJI Support for diagnosis.

There is air in the pipeline

1. Ensure that the spray tank is mounted to the aircraft properly (slightly press down the tank with the hand).
2. If the app prompts that there is air in the pipeline during the flight (and the remaining pesticide is about 1 L to 2 L), perform the following steps for troubleshooting.
3. Check whether the filter inside the spray tank is blocked. If yes, it may be causing this issue.
4. Check whether the flow meter's screw is tightened. If yes, loosen the screw, calibrate the flow meter, and tighten the screw again.



5. Replace the rubbers on the one-way valve. (It is recommended that the user separate the

spray tank from the aircraft when the aircraft is not in use. Otherwise, the rubber cannot rebound due to long-term contact. If the user does not want to replace the rubbers, adjust the rubbers' installation position. Mount rubber 1 to the lower position and rubber 2 to the upper position.



Solution: When replacing the spray tank, fill in the base with water and then mount the spray tank. At present, the passing rate of this solution is 90%.



23. There is an error with the T16 barometer

- Issue: The drone's flight height drops during hovering, or the app wrongly prompts that there is an IMU altitude error.
- Aircraft model: T16
- Solution:
 1. Ensure that the barometer holes are not blocked by debris. Then guide the user to remove the sealing rubber ring on the USB-C stopper.
 2. If the issue persists, link the remote controller with the drone and send the QR code shown in the app to DJI Support for diagnosis.

24. There is a communication error with the T16 battery and flight controller

- Issue: A warning stating that there is a communication error with the battery and flight controller is shown in the app.
- Aircraft model: T16
- Solution:
 1. Check whether or not the battery or the aircraft is malfunctioning (try to mount a functioning battery to the aircraft).
 2. If there is a battery error, check whether the issue will occur every time. If yes, return the battery to the factory. If not, check whether there is any debris on the battery port.
 3. If the drone is malfunctioning, update the drone firmware and remote controller firmware to the latest version.
 4. If the issue still cannot be fixed, check whether the power distribution board is connected well. If not, replace the board. To test the power distribution board, refer to the T16 power distribution board test guide.
 5. If the issue persists, link the remote controller with the drone and send the QR code shown in the app to DJI Support for diagnosis.

25. The data shown on the Agriculture Service Platform is inaccurate or lost

After the **MG-1P** flight records are uploaded, there is no data shown on the Agriculture Service Platform (occasionally).

Issue: The aircraft's and remote controller's firmware is V01.06. Sometimes when the flight records are uploaded and it is displayed that the upload is successful, there is no record shown on the Agriculture Service Platform.

Solution: Use DJI Assistant 2 for MG to update the aircraft firmware and remote controller firmware to the latest version.

After the **MG-1S** flight records are uploaded, there is no data shown on the Agriculture Service Platform (occasionally).

Collect the following information from the user.

1. The account and password of Agriculture Service Platform;
2. Aircraft model, aircraft's serial number, and main controller's serial number;
3. Operation area and time;
4. How many acres does the aircraft cover?
5. Check whether the operation mode is Manual or AB mode;
6. Aircraft's firmware version and remote controller's firmware version;

26. There is water or sand inside the T16 delivery pump, or the delivery pump is powerless

- Issue: There is water or sand inside the T16 delivery pump, or the delivery pump is powerless.
- Related delivery pumps for this issue: Delivery pumps that are out of warranty. For delivery pumps under warranty, disassembling the delivery pump as follows will affect its warranty repair service.
- Solution
If the delivery pump is water damaged, the connector of the cable which connects the delivery pump and spraying board might be poorly connected. In this case, remove the motor's external shell and check whether it is water damaged. Please disassemble the shell as follows:
 1. Use a 2.5 hexagonal socket screwdriver to remove the four screws securing the motor's external shell. Then remove the shell.
 2. Make the motor face downward to check whether it is water damaged. If necessary, dry the inside with a hair dryer (set to room temperature).



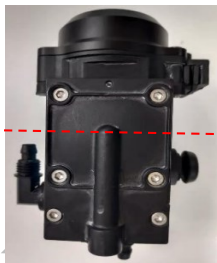
If the delivery pump is powerless, check whether the diaphragm is damaged. Please disassemble the pump as follows:

1. Use a 2.5 hexagonal socket screwdriver to remove the six screws securing the left and right pump covers. Then remove the delivery pump's cover.
2. Remove the screw and diaphragm.
3. Check whether the diaphragm is swollen or damaged. If yes, replace it.



If there is sand inside the delivery pump, please perform the following:

1. Use a 2.0 hexagonal socket screwdriver to remove the four screws securing the delivery pump valve base, then remove it.
2. Remove the two delivery pumps' one-way valve top parts first, and then remove the two one-way valve springs, as shown below.
3. Remove the one-way valve top parts and springs on the other side in the same way.
4. Clean sand or foreign objects near the water inlet and outlet of the pump.
5. You can also place the lower part of the delivery pump (the red line in the picture below) in clean water and shake it gently to let the sand fall out from the water inlet.



27. The delivery pump's motor is rotating, but the delivery pump cannot pump the water

- Cause: Since the pump is not used for a long time, the spool is stuck to the liquid or the gap is large, and the closed loop space cannot be formed.
- Aircraft model: All
- Solution
 1. Add some water to the water outlet to restart the delivery pump for a pressure closed loop.
 2. The user can use a syringe to fill the water to the inlet of the pump as shown below.



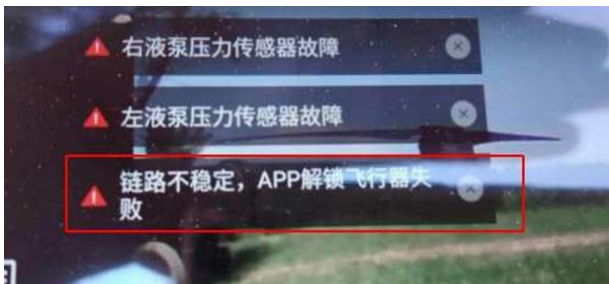
28. The drone can only take off when the RTK signal becomes stable when using RTK

- Cause: The drone can only take off when the RTK signal becomes stable when using RTK.
- Aircraft model: All
- Solution:
The drone can only take off when the RTK signal is stable and "FIX" is shown on the app.



29. A warning stating that the link is unstable is shown on the app

- Issue: A warning shown on the app states that the link is unstable and the drone has been locked by the app.



- Aircraft model: All
- Solution:
 1. Go to the image transmission settings and channel to check whether the channel mode is selected to Manual. In Manual mode, it is easy to operate at a frequency band with strong interference. Please set the Channel to Auto.



2. Go to "RC Settings" and "Set RC ID" to set a different ID for the remote controller to avoid signal interference. Then link the remote controller with the drone again.



3. Update the remote controller firmware and drone firmware to the latest version.
4. Check whether the issue is fixed after the remote controller is replaced. If yes, there is an error with the remote controller image transmission link. Ensure that the antennas are connected well. If the issue still cannot be fixed, replace the SDR module.
5. If the issue cannot be resolved after the remote controller is replaced, there is an error with the drone image transmission link. Ensure that the antennas are connected well. If the issue cannot be fixed, replace the SDR module. Replace the SDR module and main controller module (for MG-1P) or replace the aerial-electronics system module (T16).
6. If the issue persists, send the QR code shown in the app to DJI Support for diagnosis.

30. Screen replacement notice due to screen ghost touch issue

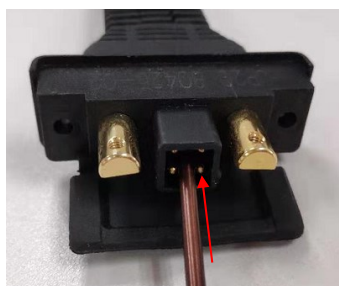
- Issue: Screen ghost touch occurs when using the remote controller.
- Related aircraft model: Phantom 4 RTK, MG-1P Series, T16
- Solution:
 1. To replace the remote controller's screen, refer to *T16 Repair Guide for Dealer*.
 2. Before replacing the screen, ensure that the remote controller's firmware version is V2.1.0012 or higher. Otherwise, the screen may be blacked out.
 3. If the issue persists after the screen is replaced, contact DJI Support.

31.The new MG-1P battery power cable cannot be attached

- Issue: The new MG-1P battery power cable cannot be attached.

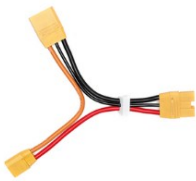


- Related accessory: MG-12000P battery power cable
- Solution:
 1. Check the cable's middle signal port and its waterproof rubber stopper. If the waterproof rubber stopper is not mounted in place, the cable cannot be attached. A specific tool is required to push the rubber stopper to the end and ensure that the rubber stopper is mounted in place.
 2. If the issue persists, contact DJI Support.



32. Aircraft + charger combinations with different batteries

Product (Aircraft + Charger)	Battery Type	Accessory
T16	T16 Battery	Normal use
MG-1S Advanced, MG-1P, MG-1P RTK	MG-12000P Battery	Normal use
	MG-12090/MG-12000S Battery	MG-1A/P Power Cable Adapter
MG-1S, MG-1	MG-12000P Battery	XT90 Shorting Plug
	MG-12000/MG-12000S Battery	Normal use
2600W 4-Channel Intelligent Battery Charger	T16 Battery	Charger (Standard)
	MG-12000P Battery	Normal use
	MG-12000/MG-12000S Battery	MG-1A/P Power Cable Adapter
MG-1S/A/P Intelligent Battery Charger 2400W MG-1S/A/P Charging Hub 1200W	MG-12000P Battery	MG-1A/P Battery Charging Balancing Cable + XT90 Shorting Plug
	MG-12000/MG-12000S Battery	Normal use
2400W DJI Hex Charger MG 2-Channel Charger 1200W MG Charger Intelligent Battery Charger (EV) 2400W	MG-12000P Battery	Normal use
	MG-12000/MG-12000S Battery	MG-1A/P Power Cable Adapter

Name	Picture
MG-1A/P Power Cable Adapter	
XT90 Shorting Plug	

MG-1A/P Battery Charging Balancing Cable	
Charger Plug	
2600W 4-Channel Intelligent Battery Charger	
MG Intelligent Battery Charger 2400W	
MG-1/S/A/P Charging Hub 1200W	
MG4-Channel Intelligent Battery Charger 2400W	
MG 2-Channel Charger 1200W	

MG Charger Intelligent Battery Charger (EV) 2400W

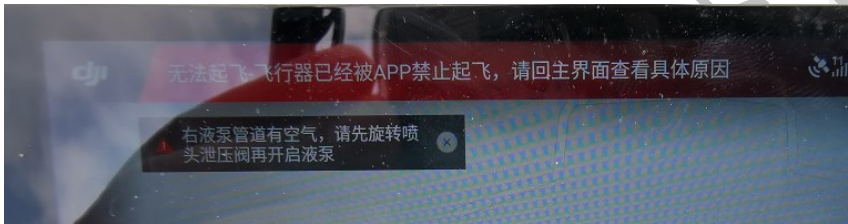


33. When the RTK station is connecting, does it need to place the drone to the pre-set calibration point to offset the bias of the flight route?

- Aircraft model: T16 and MG-1P RTK
- Solution: When the RTK station (such as D-RTK 2) is being used, Rectify Offset is required.

34. A warning stating that the drone cannot take off and is prohibited by the app is shown in the app.

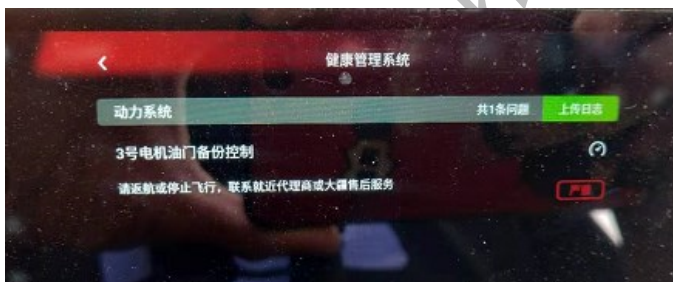
- Issue: A warning stating that the drone cannot take off and is prohibited by the app is shown in the app.



- Aircraft model: All
- 排查方案:
 1. Confirm that the user is the boss of the team; if not, please guide the customer to create a new team as a boss.
 2. If there are multiple teams under the boss, confirm that the user is logged into the correct team.
 3. Log in to DJI Agricultural Management Platform, and go to Drone Management to check whether the aircraft is locked. If it is locked, please unlock it.
 4. If it is not locked, please log out of the account and disconnect the network, then log in again.
 5. If the problem persists, please ask the customer to provide information such as the aircraft's serial number, flight controller's serial number, app error screenshots, and account number, and then send the information to DJI Support.

35.A “Motor throttle backup control” warning is shown in the app

- Issue: The app prompts a “Motor throttle backup control” warning.
- Aircraft model: T16
- Solution:
 1. Check if the corresponding ESC signal cable, power cable connector, and connector of the cable which connects the spraying board and aerial-electronics system board are loose. If yes, tighten the corresponding screw(s).
 2. If the cable is connected firmly, change the ESCs as follows: For example, if the app prompts that the ESC 1 is malfunctioning, switch the positions of ESC 1 and ESC 2 and check the app warning again. If the app then prompts that the ESC 2 is malfunctioning, it can be determined that the ESC is malfunctioning. If the app also still prompts that there is an error with the ESC 1, then the link is abnormal. Please replace the ESC cable and spraying board, and then test again.
 3. If the issue persists, link the remote controller to the aircraft and send the QR code shown in the app to DJI Support for diagnosis.



36.The flight mode of the Phantom 4 RTK does not match the command of the remote controller flight mode switch

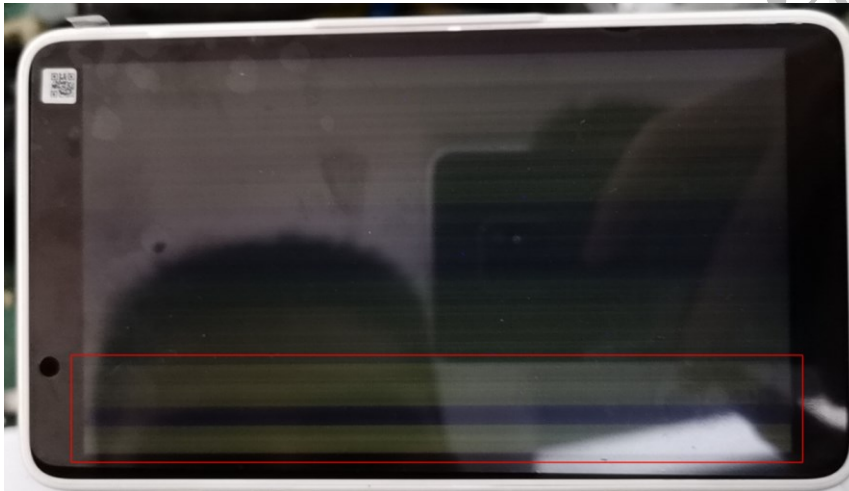
- Issue: When the flight mode switch on the remote controller is toggled to P, the app prompts that the drone is in ATTI mode. When the flight mode switch on the remote controller is toggled to A, the app prompts that the drone is in GPS mode.
- Aircraft model: Phantom 4 RTK and remote controller with built-in screen
- Solution:
 1. Check whether the drone was controlled by the SDK remote controller (without built-in screen). If yes, link the SDK remote controller with the drone again and turn off Multiple Flight Modes in the app.
 2. If not, ask the user to provide the video and flight records to DJI Support.

37. The app on the MG series remote controller prompts that there is a communication error between the

- Issue: The app on the MG series remote controller prompts that there is a communication error between the Intelligent Battery and flight controller.
- Aircraft model: MG Series
- Solution:
 1. Check the battery XT100 connector's 4-pin link which transmits the battery encryption information to the main controller.
 2. Link: Battery Communication Cable – Aircraft XT100 Connector Module – Power Distribution Board Module – Central Board – CAN Cable Connecting Central Board and Main Controller – Main Controller. Troubleshoot each part of the link.
 3. Try to replace the battery. If the issue is fixed, mount the defective battery to another drone and check if the issue occurs. If yes, there is an error with the battery communication cable.
 - a. If it is an MG-12000P battery, replace the battery communication cable.
 - b. If it is an MG-12000 or MG-12000S battery, it is recommended that the user send back the battery for repair.
 4. Remove the delivery pump and disconnect the cable connecting the radar and central board. If the issue is then fixed, the delivery pump or radar is water damaged and short circuited.
 5. If the issue persists, replace the XT100 port.
 6. If it is an MG-1S drone, replace the CAN port on the central board. Ensure that the CAN cable's silver metal side faces downwards. As there is only one CAN port on the MG-1A and MG-1P, there is no need to replace the CAN port.
 7. Replace the central board and update the firmware again. If the central board is damaged, the app will prompt that the delivery pump and radar are not connected.
 8. If the issue persists after the central board is replaced, check whether the external barometer is removed (for MG-1S only; there is no external barometer for the MG-1A and MG-1P).
 9. If the issue still cannot be resolved, replace the power distribution board and XT100 connector.
 10. If the issue persists, export the corresponding data and contact DJI Support.

38. Stripes appear after the remote controller screen is replaced

- Issue: Stripes appear after the remote controller screen is replaced.



- Aircraft model: Phantom 4 RTK, MG-1P Series, and T16
- Solution
 1. Check whether the firmware is V2.1.0012 or higher before replacing the screen.
 2. Connect the remote controller to a PC and update the firmware using DJI Assistant 2 For MG.
 3. If the issue persists, contact DJI Support.

39. Difference between the empty tank warning occurring when there is little liquid and when the spray tank is empty

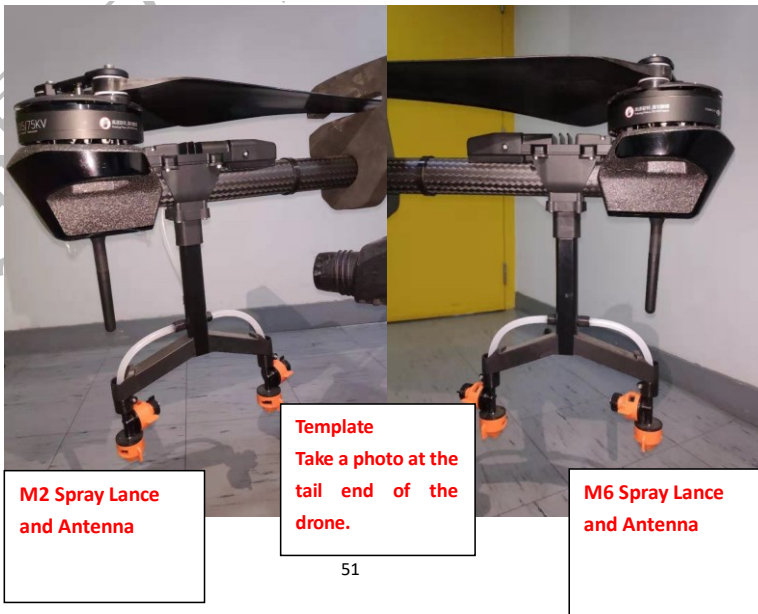
- Aircraft model: All
- Difference:
 1. In No Remaining mode, the empty warning will be displayed when the tank is nearly empty.
 2. In Little Remaining mode, the empty warning will be displayed when there is about 300 ml remaining.

40. Photos of T16 is required after crash

- Background: After the drone crashes, relevant pictures of the drone is required for data analysis to facilitate faster and more accurate positioning of the problem, as well as to avoid delays in communication between users and DJI.
- Aircraft model: T16
- Requirements:
 1. Unfold the aircraft arms and take a photo at the tail end of the drone. Ensure that the motors, propellers, and spray lances are clear.



2. Stand behind the tail of the drone and take photos of the M2 and M6 motors, spray lances, and antennas to show the damage condition.



3. Stand in front of the drone's nose and take photos of the M3 and M5 propellers to show the damage condition.



4. If the aircraft is repaired in advance, please keep the relevant old components for later troubleshooting, such as whether there is any debris on the components or if there are unauthorized modifications to the product.
5. If there are any problems, contact DJI Support.

41. DJI Agras drone new and old components

Compatibility List						
Component	MG-1	MG-1S (RTK)	MG-1S/A	MG-1P (RTK)	T16	Remarks
ESC	Exclusive	Compatible			Exclusive	
Motor	Compatible				Exclusive	Note that the propellers must be used in pairs.
Propellers	Compatible (Foldable propellers 21/21R)				Exclusive (Foldable propellers R3390)	
Small Arm	Exclusive	Compatible			Exclusive	The length of the MG-1P small arm is 5 mm shorter than that of the MG-1S, but they are interchangeable.
Large Arm	Exclusive	Compatible			Exclusive	
Landing Gear Bottom Tube	Compatible				Exclusive	The length of the MG-1P bottom tube is shorter than that of the MG-1S, but they are interchangeable.
Landing Gear Crossbeam	Exclusive	Compatible			Exclusive	
GPS/RTK Antenna	Exclusive	Exclusive	Exclusive	Exclusive	Exclusive	
Central	Exclusive	Exclusive	Compatible		Exclusive	

Board	ve				
Antenna	Exclusive	Exclusive	Compatible	Exclusive	
Nozzle	Compatible				
Spray Lance	Exclusive	Compatible	Exclusive		
Delivery Pump	Exclusive	Exclusive	Compatible	Exclusive	
Spray Tank	Exclusive	Exclusive	Compatible	Exclusive	The MG-1S radar cannot be mounted to the new MG-1P spray tank.
Remote Controller	Exclusive	Exclusive	Exclusive	Compatible	From firmware V02.01.0009, the Phantom 4 RTK's, T16's, and MG-1P (RTK)'s remote controllers are interchangeable.

42. Nozzle model and spray rate

MG-1S Nozzle Spray Rate			
Nozzle Model	XR11001VS	XR110015VS	TX-VK08
Spray Rate (L/min)	0.64~0.9	0.96~1.3	0.78~1.2
	1.28~1.8	1.92~2.6	1.56~2.4

MG-1P Nozzle Spray Rate						
Nozzle Model	XR11001VS	XR110015VS	TP1100050	TX-VK04	TX-VK06	TX-VK08
Spray Rate (L/min)	0.64~0.9	0.96~1.3	0.32~0.46	0.44~0.6	0.66~0.9	0.78~1.2
	1.28~1.8	1.92~2.6	0.64~0.92	0.88~1.2	1.32~1.8	1.56~2.4

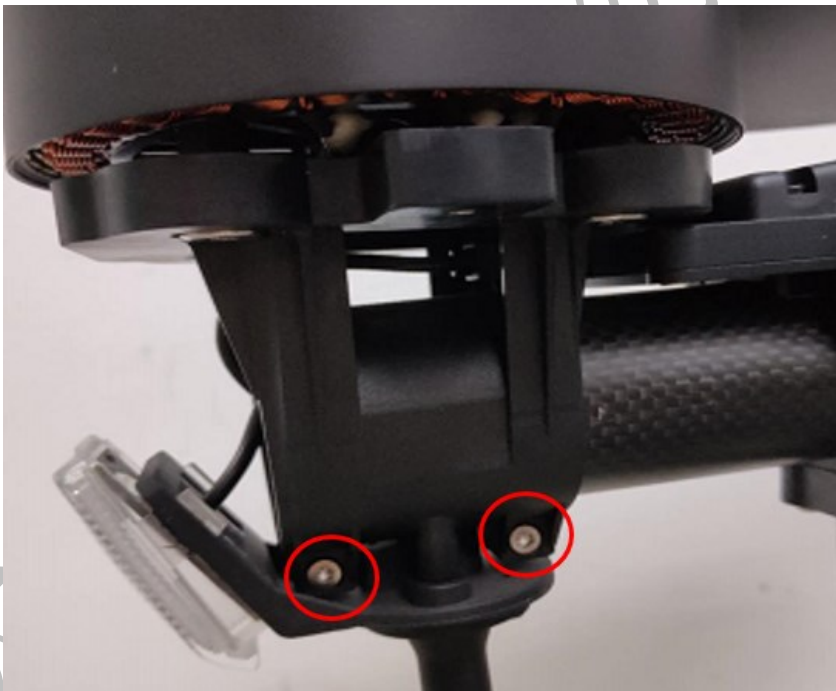
T16 Nozzle Spray Rate		
Nozzle Model	XR11001VS	XR110015VS
Spray Rate (L/min)	1.28~1.8	1.92~2.4
	2.56~3.6	3.84~4.8

Note: The data on the first row is when half of the delivery pumps are open, and the data on the second row is when all pumps are turned on.

43. The T16 spins during flight

- Issue: The T16 spins during the flight.
 - The issue occurs because there is an error with the motor angle. Please check if the motor base is not tightened and if the limit part is rubbed off.
1. If the motor base is loose but the limit part is not rubbed off, apply the thread locker to the screws securing the aircraft arm.
 2. If the limit part is rubbed off, replace the motor base and apply the thread locker.
 3. If the issue persists, send the QR code to DJI Support for diagnosis.
 4. It is recommended that the repair staff check whether the motor base is loose during the repair. If yes, apply the thread locker.

Position of screws on the motor base:

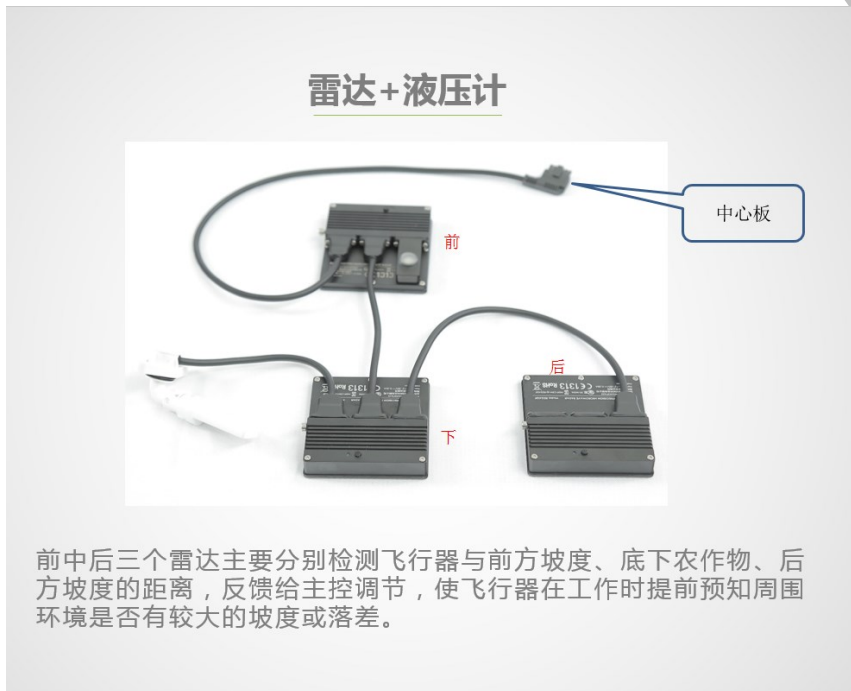


Position of limit part on the motor base



44. MG-1S Radar LED Settings

The front radar on the aircraft nose is solid blue, the middle radar is solid yellow, and the rear radar on the tail of the drone is solid green.



1. Mount the battery to the drone and power it on.
2. Press and hold the black button beside the radar LED, then connect the radar to the drone.



3. When the LED blinks yellow, green, blue, and white alternatively, and then it becomes solid white, release the button. Then press the button again to set the LED color.



45. Some areas around corners are not sprayed by the T16 (as shown by the white lines in the figure below)



- Aircraft model: T16
- Cause: The calibration data is wrong, causing the delivery pump to not rotate promptly and the flow rate to be low.
- Solution:
 1. Calibrate the delivery pump. For more details, visit <https://www.skypixel.com/videos/t16-0bf01265-a50d-433f-8254-82dab021aa7a>.
 2. To check whether the calibration is successful, toggle the spray rate dial and press the spray button to start spraying. After the water volume is stable, the maximum flow rate is 3.4 L/min - 3.8 L/min, and the minimum flow rate is 1.16 L/min - 1.4 L/min (original nozzle).
 3. If the issue persists, link the remote controller with the drone, export the QR code, and send it to DJI Support.

批注 [k4]: 提供英文链接

46. How far away from high-voltage towers/high-voltage wires can the T16 work normally?

The distance requirements in different modes are as follows:

1. GNSS Mode: 20 m and above
2. RTK Mode: 10 m and above

To avoid interference from high-voltage wires to the T16 compass, it is recommended to turn on the RTK mode as much as possible when working near any high-voltage lines for safe operation.

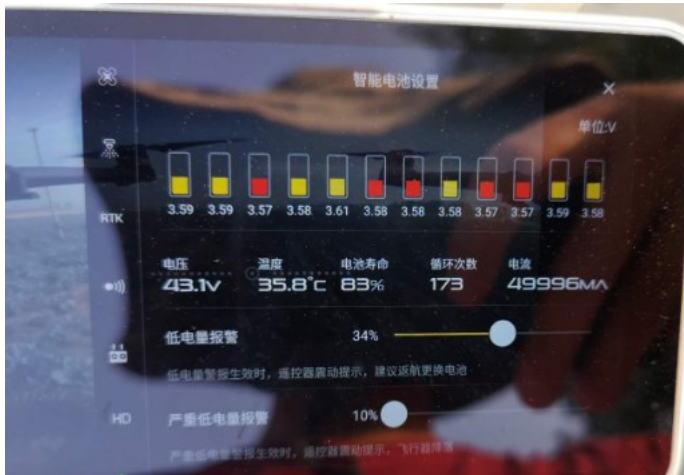
47. T16 M1, M2, M3, M4, M5, and M6 motors' angles and rotation directions

1. M4 motor: 0° (foolproof design)
2. M3, M5, and M6 motors: 5° (foolproof design)



48. The MG/T16 battery cells' colors shown in the app are different

Issue: The MG/T16 battery cells' colors shown in the app are different.



The battery cell's color will change based on the current voltage.

One battery cell's color changes as follows:

1. MG-1P: When the voltage is larger than 3.67V, the color is green; when the voltage is 3.58-3.67V, the color is yellow; and when the voltage is lower than 3.58V, the color is red.
2. T16: When the voltage is larger than 3.5V, the color is green; when the voltage is 3.5-3.42V, the color is yellow; and when the voltage is lower than 3.42V, the color is red.
3. If the color is red or a critical low battery level warning is shown in the app, please charge the battery promptly.

49. MG/T16 Battery Times Charged

Calculation Logic

1. The T16 battery capacity is 17500 mAh. When the battery discharges (consumes) 75%, one time charged will be counted.
2. The MG series battery times charged logic is the same as the T16: 75% discharge (consumed) will be calculated as one time charged.
3. For example, if the battery consumes 50% of its power after fully charged, then is fully charged again, and then consumes 25% again, then it will be calculated as one time charged.

50. How does the DJI Hex Charger charge two batteries simultaneously?

- The user can choose between two different charging modes: quick charge and slow charge. The logic of the two modes is the same, but the charging time is different. DJI Hex Charger can charge two batteries at the same time (one from each side). If the two batteries are placed on the same side, the charger will charge the batteries in order of their power level, starting from highest to lowest. If the batteries are placed on different side, the charger will charge the battery with the lowest power level. Once all batteries have the same power level, they will all be charged simultaneously.
- The charger can charge different types of batteries. For example, one MG-1P battery and one MG-1S battery can be charged simultaneously.

51. How long does it take to fully charge the MG series remote controller? How long can the remote controller work?

- Aircraft model: MG-1S, MG-1S Advanced, MG-1P, T16, and Phantom 4 RTK
 1. It takes 3.5 hours or 4 hours to fully charge the MG-1A remote controller (Chinese version), and it works for about 5 hours.
 2. It takes 3 hours to fully charge the MG-1A remote controller (Japanese version), and it works for about 4 hours.
 3. It takes 1 hour to fully charge the MG-1P/T16/Phantom 4 RTK remote controller (single battery), and it works for about 3 hours.
 4. It takes 4 hours to fully charge the MG-1S remote controller, and it works for about 5 hours.

52. Can the MG-1S remote controller be used to control the MG-1A?

1. The MG-1S remote controller be used to control the MG-1A, but some functions are not available.
2. The MG-1A remote controller cannot be used to control the MG-1S, as some functions are unavailable, and some may be unstable.
3. The MG-1S is not allowed to be used with the MG-1A remote controller.

53. MG Series GEO Zones Range Inquiry

The GEO zones for the MG drones are circular areas. You can check the GEO map by visiting <https://www.dji.com/cn/flysafe/geo-map>.

54. When the MG drone is not in a GEO zone, but a warning shown in the app states that the drone is in a warning zone and cannot take off

1. There is an error with the remote control cache. Please guide the user to clear the remote control cache.
2. If the issue persists, export the flight controller data to the DJI After-sales department for diagnosis.

55. Can the MG-1P RTK be used with the MG-1S RTK base station and handheld mapper?

No. The MG-1P RTK can only be used with a new base station and D-RTK handheld mapper.

56. Does the T16's 2600W 4-channel charger need to connect to a power supply for firmware updates?

There is no relation between the T16's 4-channel charger and the power source. The charger firmware can be updated by using DJI Assistant 2 no matter whether the power source is connected or not.

57. What do the antenna 1 and antenna 2 shown on the T16 DJI MG2 app refer to?

The antenna 1 and antenna 2 shown on the app are as shown below:



58. How to check whether the T16 drone's motor triple-phase line is disconnected

Method:

After disconnecting the motor from the ESC, use a multimeter to test the resistance between the two terminals of the motor. For example, use the multimeter to touch the terminal A and the terminal B respectively, and select the appropriate range to read the A-B phase resistance. Then the A-C phase resistance and the B-C phase resistance can also be measured.

When to check: It is recommended to perform this check after the drone has crashed, the app prompts that the motor is obstructed, or if the motor temperature is abnormal.

Standards: The resistance of A-B, B-C, and A-C is $80 \pm 3.2 \text{ m}\Omega$.



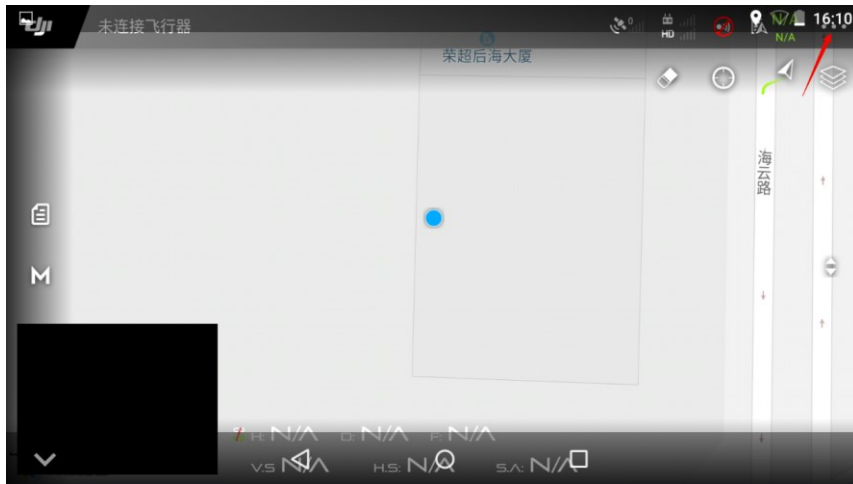
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59. After the field planned by the Phantom 4 RTK using DJI Terra is uploaded to the DJI MG2 app for the T16, the map is not displayed, and only the flight route can be shown

- The map needs to be set in the DJI MG2 app when the field planned by the Phantom 4 RTK is uploaded to the SD card.
 1. Tap "Execute Operation".



2. Tap "•••" to enter More settings.



3. Go to General Settings.



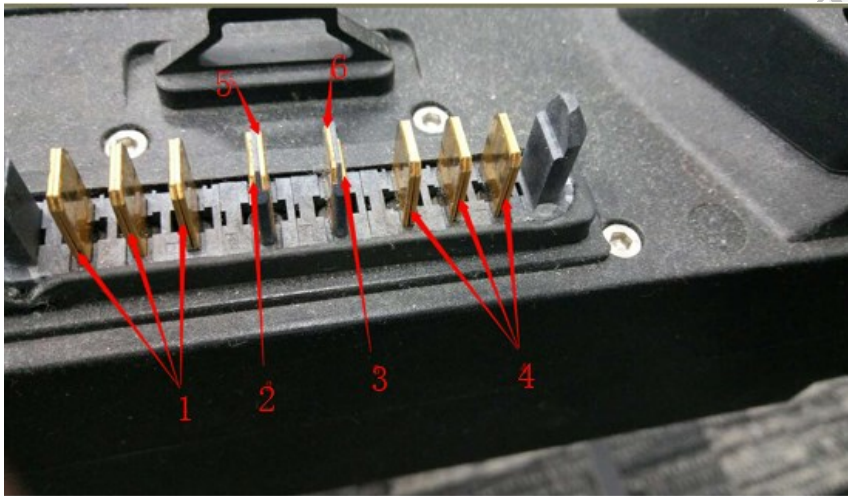
4. Turn on "Choose HD Image Scheme".



5. Go back to the Home screen and tap “Execute Operation” again, and the map will be displayed normally.

60. T16 Power Distribution Board Test Guide

- Issue: When a warning is shown in the app that states that there is an error with the battery communication or the power distribution board is abnormal, perform the following to determine whether the power distribution board is malfunctioning.
- Power distribution board Pin definition:



Pin1:14S- Pin2: VCC5V Pin3:VCC5V_GND Pin4:14S+ Pin5: 485A Pin6:485B

- Solution:

Check the appearance of the power distribution board to see if there are any pesticide residues. If yes, clean them up promptly and check whether the drone's power circuit is short-circuited.

1. Use a multimeter to check whether a short circuit occurs between Pin1 and Pin4. If yes, it means that the drone's power cable is short-circuited. Then check whether the drone is water damaged, whether a short circuit occurred if the drone hit a high-voltage line, or whether the power cable of the ESC is damaged or short-circuited.
2. Check whether a short circuit occurs between Pin1 and Pin2. If yes, the power distribution board is damaged. Replace it.
3. Check whether a short circuit occurs between Pin2 and Pin5. If yes, the power distribution board is damaged. Replace it.
4. Check whether a short circuit occurs between Pin5 and Pin6. If yes, the power distribution board is damaged. Replace it.
5. Check whether a short circuit occurs between Pin3 and Pin6. If yes, the power distribution board is damaged. Replace it.
6. Check whether a short circuit occurs between Pin3 and Pin4. If yes, the power distribution board is damaged. Replace it.

7. Use the multimeter to measure resistance between Pin1 and Pin3. If it is less than 3 ohms, the power distribution board VCC5V_GND connection is normal. Otherwise, the power distribution board is malfunctioning and needs to be replaced.